ARATA HORIE

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EDUCATION

The University of Tokyo — **Tokyo, Japan**
Ph.D. in Engeneering — *April 2019 - March 2022*
Department of Advanced Interdisciplinary Studies
Advisor: Dr. Masahiko Inami

Tohoku University — **Sendai, Japan**
M.S. in Information Science — *April 2017 - March 2019*
Department of Applied information science
Advisor: Dr. Masashi Konyo

Tokyo University of Science — **Tokyo, Japan**
B.S. in Mechanical Engineering — *April 2013 - March 2017*
Department of Mechanical Engineering
Advisor: Dr. Hiroshi Mizoguchi

WORK EXPERIENCE

commissure Inc. — **Tokyo, Japan**
CTO — *January 2023 - present*

Keio University Graduate School of Media Design, Embodied Media — **Tokyo, Japan**
Project Senior Assistant Professor — *April 2026 - present*

Keio University Graduate School of Media Design, Embodied Media — **Tokyo, Japan**
Project Assistant Professor — *April 2023 - March 2026*

The University of Tokyo, Research Center for Advanced Science and Technology, Information Somatics Lab — **Tokyo, Japan**
Project Assistant Professor — *April 2022 - March 2023*

Information Somatics Lab, The University of Tokyo — **Tokyo, Japan**
Research Assistant / JSPS Research Fellow DC2, Advisor: Masahiko Inami — *April 2019 - March 2022*

exiii — **Tokyo, Japan**
Haptic Research Internsip — *April 2019 - September 2019*

Human-Robot Informatics Lab, Tohoku University — **Sendai, Japan**
Research Assistant, Advisor: Masashi Konyo — *April 2017 - March 2019*

GRANTS

ACT-X Hardware in Future for Resilience of Real Space, "Design of Self-other Inseparable Intercorporeality Through Body Surface Deformation Device", JST — **2021 - 2023**
Japan Science and Technologies

Japan Society for Promotion of Science DC2, "Self-motion Sensation Induction by the Presentation of Distributed Force", JSPS — **2020 - 2021**
Japan Society for Promotion of Science

HONORS AND AWARDS

Audience Choice Award, SIGGRAPH 2026 Emerging Technologies **2026**
Co-author, "EmoMime: Augmenting Social Behavior and Self-Expression via Wearable Robotic Limbs"

Best Paper Award, CHI 2025 **2025**
Co-author, ""*It' s Like Being On Stage*"* *: Conveying Dancers' Expressiveness Through A Haptic-Installed Contemporary Dance Performance"*

Best Demo Award, SIGGRAPH Asia 2025 Emerging Technologies **2025**
Co-author, "Experience Sharing Box: A Platform for Capturing, Storing, and Playback of Multisensory Memory"

Official Selection for Laval Vitual, SIGGRAPH 2024 Emerging Technologies **2024**
Co-author, "FEELTECH Wear: Enhancing Mixed Reality Experience with Wrist to Finger Haptic Attribution"

Best Demo Award, EuroHaptics Conference 2024 **2024**
Co-author, "Rotational Skin-stretch Distribution Creates Directional Force Sensation on the Wrist"

Dean's Award, Graduate School of Engineering **2022**
The University of Tokyo

Fellowships from Japan Society for Promotion of Science DC2, JSPS **2020 - 2021**
Japan Society for Promotion of Science

Best Paper Award, UbiComp MIMSVAI 2021 **2021**
Co-author, "High-speed non-contact thermal display using infrared rays and shutter mechanism"

Best Paper Award Nominated, World Haptics Conference 2021 **2021**
Author, "Two-dimensional moving phantom sensation created by rotational skin stretch distribution"

Outstanding Performance Award, Asia Digital Art Award 2020 **2020**
Author, "TorsionCrowds"

Disruptive Challenge Finalist, INNO-Vation 2019 **2019**
Author, Ministry of Internal Affairs and Communications in Japan

Best Demonstration Award Gold Winner, AsiaHptics 2018 **2018**
Author, "Enhancing Haptic Experience in a Seat with Two-DoF Buttock Skin Stretch"

Ultrahaptics Student Challenge Finalist, EuroHaptics 2018 **2018**
Co-author, "Weather Teleport"

Robomech Award, Robotics and Mechatoronics 2018 **2018**
Author, "Induction of acceleration sensation of self-motion by presenting shear force to buttocks"

Young Researcher's Award, VRSJ 2018 **2018**
Author, "Presenting a pushing up feeling from the seat by buttocks skin stretch"

Excellent Presentation Award, SICE SI 2018 **2018**
Author, "2-DoF Buttock Skin Stretch for Inducing Acceleration Perseption of Self-motion"

COLOPL Award, International collegiate Virtual Reality Contest 2017 **2017**
Team Manager, "Shall we coffee cup?"

Young Researcher's Award, SICE SI 2017 **2017**
Author, "Sliding direction and speed presentation Combining high frequency and asymmetric vibration"

Exellent Presentation Award, SICE SI 2017 **2017**
Author, "Sliding direction and speed presentation Combining high frequency and asymmetric vibration"

PEER-REVIEWED PUBLICATIONS

JOURNAL PAPER

1. Arata Horie, Hideki Shimobayashi, Hiroto Saito, and Masahiko Inami. Designing distributed-type haptic device based on rotational skin stretch. *Transactions of the Virtual Reality Society of Japan*, 25(4):402–411, 2020

2. Hiroto Saito, Arata Horie, Azumi Maekawa, Seito Matsubara, Sohei Wakisaka, Zendai Kashino, Shunichi Kasahara, and Masahiko Inami. Transparency in human-machine mutual action. *Journal of Robotics and Mechatronics*, 33(5):987–1003, 2021. **(Co-first author)**

CONFERENCE PAPER

1. Arata Kotani and Arata Horie. Audiohapticdiffusion: Synchronized audio-haptic generation via shared latent space. In *EuroHaptics 2026*, volume 16593 of *Lecture Notes in Computer Science*, pages 151–160, Cham, 2026. Springer. **(Co-first author)**

2. Arata Kotani, Hideki Shimobayashi, and Arata Horie. Ders: Distribution of eccentric rotational skin-stretch induce directional force sensation on a wrist. In *EuroHaptics 2026*, volume 16593 of *Lecture Notes in Computer Science*, pages 249–258, Cham, 2026. Springer. **(Co-first author)**

3. Emiri Yorioka, Yufan Zhu, Jinsuke Morita, Takatoshi Yoshida, Sohei Wakisaka, Arata Horie, and Kouta Minamizawa. Piano duo: Haptic sharing system supports two-piano practice. In *2025 IEEE World Haptics Conference (WHC)*, pages 1–6. IEEE, 2025

4. Ximing Shen, Xuan Li, Youichi Kamiyama, Danny Hynds, Giulia Barbareschi, RAY Lc, Sohei Wakisaka, Arata Horie, and Kouta Minamizawa. " it's like being on stage": Conveying dancers' expressiveness through a haptic-installed contemporary dance performance. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*, pages 1–18, 2025**(Best Paper Award)**

5. Rodan Umehara, Arata Horie, and Kouta Minamizawa. Rotational skin-stretch distribution creates directional force sensation on the wrist. In *International Conference on Human Haptic Sensing and Touch Enabled Computer Applications*, pages 238–250. Springer, 2024

6. Arata Horie, Yunao Zheng, and Masahiko Inami. A wearable system integrating force myography and skin stretch feedback toward force skill learning. In *2023 IEEE World Haptics Conference (WHC)*, pages 190–196. IEEE, 2023

7. Ryo Murata, Arata Horie, and Masahiko Inami. Dynamic derm: Body surface deformation display for real-world embodied interactions. In *Proceedings of the Augmented Humans International Conference 2023*, AHs '23, page 267 – 277, New York, NY, USA, 2023. Association for Computing Machinery

8. Sosuke Ichihashi, Arata Horie, Masaharu Hirose, Zendai Kashino, Shigeo Yoshida, and Masahiko Inami. High-speed non-contact thermal display using infrared rays and shutter mechanism. UbiComp '21, page 565 – 569, New York, NY, USA, 2021. Association for Computing Machinery**(Best Paper Award)**

9. Arata Horie, Zendai Kashino, Hideki Shimobayashi, and Masahiko Inami. Two-dimensional moving phantom sensation created by rotational skin stretch distribution. In *2021 IEEE World Haptics Conference (WHC)*, pages 139–144, 2021**(Nominated to Best Paper Award)**

10. Arata Horie, MHD Yamen Saraiji, Zendai Kashino, and Masahiko Inami. Encounteredlimbs: A room-scale encountered-type haptic presentation using wearable robotic arms. In *2021 IEEE Virtual Reality and 3D User Interfaces (VR)*, pages 260–269. IEEE, 2021

11. Hideki Shimobayashi, Tomoya Sasaki, Arata Horie, Riku Arakawa, Zendai Kashino, and Masahiko Inami. Independent control of supernumerary appendages exploiting upper limb redundancy. In *Augmented Humans Conference 2021*, AHs'21, page 19 – 30, New York, NY, USA, 2021. Association for Computing Machinery

12. Yudai Tanaka, Arata Horie, and Xiang 'Anthony' Chen. Dualvib: Simulating haptic sensation of dynamic mass by combining pseudo-force and texture feedback. In *26th ACM Symposium on Virtual Reality Software and Technology*, VRST '20, New York, NY, USA, 2020. Association for Computing Machinery

13. Arata Horie, Hikaru Nagano, Masashi Konyo, and Satoshi Tadokoro. Buttock skin stretch: Inducing shear force perception and acceleration illusion on self-motion perception. In *International Conference on Human Haptic Sensing and Touch Enabled Computer Applications*, pages 135–147. Springer, 2018

DEMO

1. Hideki Shimobayashi, Tomoya Sasaki, Masaharu Hirose, Arata Horie, and Masahiko Inami. Emomime: Augmenting social behavior and self-expression via wearable robotic limbs. In *ACM SIGGRAPH 2026 Emerging Technologies*, pages 1–3. 2026 **SIGGRAPH 2026 Emerging Technologies Audience Choice Award**

2. Haruomi Higashi, Yufan Zhu, Kohei Kobayashi, Kiryu Tsujita, Jinsuke Morita, Emi Koizumi, Jiayi Shen, Mauricio Sousa, Arata Horie, Takatoshi Yoshida, and Kouta Minamizawa. Experience sharing studio: An immersive platform for spatio-temporal recording and playback of embodied multisensory memory. In *ACM SIGGRAPH 2026 Emerging Technologies*, pages 1–3. 2026

3. Haruomi Higashi, Yufan Zhu, Jack Brophy, Ismael Rasa, Jinsuke Morita, Arata Horie, Takatoshi Yoshida, and Kouta Minamizawa. Experience sharing box: A platform for capturing, storing, and playback of multisensory memory. In *Proceedings of the SIGGRAPH Asia 2025 Emerging Technologies*, pages 1–2. 2025 **SIGGRAPH Asia 2025 Emerging Technologies Best Demo Award**

4. Rodan Umehara, Harunobu Taguchi, Arata Horie, Yusuke Kamiyama, Shin Sakamoto, Hironori Ishikawa, and Kouta Minamizawa. Feeltech wear: Enhancing mixed reality experience with wrist to finger haptic attribution. In *ACM SIGGRAPH 2024 Emerging Technologies*, pages 1–2. 2024 **SIGGRAPH 2024 Emerging Technologies Official Selection for Laval Vitual**

5. Kiryu Tsujita, Takatoshi Yoshida, Kohei Kobayashi, Arata Horie, Nobuhisa Hanamitsu, and Kouta Minamizawa. Haptoroom: Using vibrotactile floor interfaces to enable reconfigurable haptic interaction onto any furniture surfaces. In *ACM SIGGRAPH 2024 Emerging Technologies*, pages 1–2. 2024

6. Rodan Umehara, Arata Horie, and Kouta Minamizawa. Rotational skin-stretch distribution creates directional force sensation on the wrist. In *International Conference on Human Haptic Sensing and Touch Enabled Computer Applications*, pages 238–250. Springer, 2024 **Best Demo Award**

7. Harunobu Taguchi, Youichi Kamiyama, Kenta Kan, Yulan Ju, Arata Horie, Yoshihiro Tanaka, Hironori Ishikawa, and Kouta Minamizawa. Multichannel haptic communication platform with wearable sensing and display. In *SIGGRAPH Asia 2023 Emerging Technologies*, pages 1–2. 2023

8. Aoi Uyama, Youichi Kamiyama, Sohei Wakisaka, Arata Horie, Tatsuya Saito, and Kouta Minamizawa. Somatic music: Enhancing musical experiences through the performer's embodiment. In *SIGGRAPH Asia 2023 Posters*, pages 1–2. 2023

9. Arata Horie, Ryo Murata, Zendai Kashino, and Masahiko Inami. Seeing is feeling: A novel haptic display for wearer-observer mutual haptic understanding. In *ACM SIGGRAPH Asia 2022 Emerging Technologies*, SIGGRAPH Asia '22, New York, NY, USA, 2022. Association for Computing Machinery

10. Sosuke Ichihashi, Arata Horie, Masaharu Hirose, Zendai Kashino, Shigeo Yoshida, Sohei Wakisaka, and Masahiko Inami. Thermoblinds: Non-contact, highly responsive thermal feedback for thermal interaction. In *ACM SIGGRAPH 2022 Emerging Technologies*, SIGGRAPH '22, New York, NY, USA, 2022. Association for Computing Machinery

11. Arata Horie, Hideki Shimobayashi, and Masahiko Inami. Torsioncrowds: Multi-points twist stimulation display for large part of the body. In *ACM SIGGRAPH 2020 Emerging Technologies*, SIGGRAPH '20, New York, NY, USA, 2020. Association for Computing Machinery

12. Arata Horie, Masashi Konyo, Hikaru Nagano, and Satoshi Tadokoro. Whole-body perception of ground geometry by buttock skin stretch. In *IEEE World Haptics Conference*. IEEE, 2019

13. Arata Horie, Akito Nomura, Kenjiro Tadakuma, Masashi Konyo, Hikaru Nagano, and Satoshi Tadokoro. Enhancing haptic experience in a seat with two-dof buttock skin stretch. In *International AsiaHaptics conference*, pages 134–138. Springer, 2018 **Best Demo Award Gold Winner**

14. Arata Horie, Hikaru Nagano, Masashi Konyo, and Satoshi Tadokoro. Buttock skin stretch: Inducing shear force perception and acceleration illusion on self-motion perception. In *International Conference on Human Haptic Sensing and Touch Enabled Computer Applications*, pages 135–147. Springer, 2018

WHITE PAPERS

1. Giulia Barbareschi, Ximing Shen, Yulan Ju, Diego Compagna, Frank O. Flemisch, Kenta Hashiura, Hsin-Ni Ho, Arata Horie, Keigo Inukai, Jieun Kim, Nicole Krämer, Takumi Kuhara, Sven Martin, Stephanie M. Mueller, Aiko Murata, Gernot Oberfell, Mark Paterson, Mina Shibasaki, Anna Spiegel, Celia Spooden, Heiko Stoff, Rikako Sumida, Yoshihiro Tanaka, Sohei Wakisaka, Junji Watanabe, Jan Wertel, Nora Kottmann, Kouta Minamizawa, and Sakura Yamamura. Moving forward with mediated social touch - a multidisciplinary and multicultural perspective: A white paper proposing considerations from technical, social and psychophysical disciplines for technology-mediated touch interactions encompassing german and japanese contexts. White paper, RWTH Aachen University, Aachen, Germany, 2025

NON-REVIEWED PUBLICATIONS (FIRST AUTHOR)

1. Arata Horie and Masahiko Inami. Basic study of rotating tactors array presenting strain distribution on back skin. In *Annual Conference of the Virtual Reality Society of Japan*. VRSJ, 2019

2. Arata Horie, Akito Nomura, Kenjiro Tadakuma, Hikaru Nagano, Masashi Konyo, and Satoshi Tadokoro. 2-dof buttock skin stretch for inducing acceleration perseption of self-motion. In *SICE System Integration*, volume 2018. SICE, 2018 **Excellent Presentation Award**

3. Arata Horie, Hikaru Nagano, Masashi Konyo, and Satoshi Tadokoro. Presenting pushing up feeling from the seat by buttocks skin stretch. In *Annual Conference of the Virtual Reality Society of Japan*, volume 2018. VRSJ, 2018 **Young Researcher's Award**

4. Arata Horie, Hikaru Nagano, Masashi Konyo, and Satoshi Tadokoro. Induction of acceleration sensation of self-motion by presenting shear force to buttocks. In *Robotics Mechatronics Conference 2018*. JSME, 2018 **Robomech Award**

5. Arata Horie, Hikaru Nagano, Masashi Konyo, and Satoshi Tadokoro. -2nd report- sliding direction and speed presentation combining high frequency and asymmetric vibration. In *SICE System Integration*, volume 2017. SICE, 2017 **Excellent Presentation Award, Young Researcher's Award**

6. Arata Horie, Hikaru Nagano, Masashi Konyo, and Satoshi Tadokoro. Sliding direction and speed presentation combining high frequency and asymmetric vibration. In *Annual Conference of the Virtual Reality Society of Japan*, volume 2017. VRSJ, 2017

PATENT

力覚提示装置及び力覚提示方法 **日本特許第 7313079 号 / 米国特許第 11947725 号**
堀江新（*75/100*）, 稲見昌彦（*25/100*）

力覚提示装置及び力覚提示方法 **日本特許第 7627059 号**
堀江新（*75/100*）, 稲見昌彦（*25/100*）

力覚提示装置、運動教示装置、力覚提示方法、運動教示方法およびプログラム **特開 2025-002797**
堀江新（*75/100*）, 稲見昌彦（*25/100*）

運動教示装置、運動教示システム、運動教示方法およびプログラム **WO2022/145258**
堀江新（*45/100*）, 檜山敦（*35/100*）, 稲見昌彦（*20/100*）

TEACHING EXPERIENCE

Otherness and Interaction Technology in Embodied Interaction
Keio University Graduate School of Media Design
Guest Lecturer *November 2023 - Present*

Digital Interaction Prototyping
Keio University Graduate School of Media Design
Lecturer *May 2023 - Present*

Data Processing
Tohoku University
Teaching Assistant *October 2017 - February 2018*